ShellArc Railgun Technical Memo

MiDUki

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1 Introduction

This is a mathematical model designed for schedule and production management in a 2D animation production pipeline.

2 Definition of the Three-Dimensional Domain Used to Demonstrate Progress

Define a discrete three-dimensional lattice space Γ, in which each discrete point is defined as $(x,\, y,\, z)$, where

$$\begin{cases} x \in [1,\, c_1] \cap \mathbb{Z}_+ & (c_1 \in (1, \infty) \cap \mathbb{Z}_+) \\ y \in [0,\, F] \cap \mathbb{Z} & (F \in (1, \infty) \cap \mathbb{Z}_+) \\ z \in [0,\, c_2] \cap \mathbb{Z}_+ & (c_2 \in (1, \infty) \cap \mathbb{Z}_+) \end{cases} \tag{1}$$

Here c_1, F, and c_2 denote the total cut count, the component-division fineness, and the project total length (day count), respectively.

Submissions are represented by a group of points defined as

$$\begin{cases} M = \left\{\, p_k = (x_k,\, y_k,\, z_k) \mid k = 1, 2, \ldots, |M| \,\right\} \cap \Gamma \\ \forall\, (x, y) \in \mathbb{Z}^2,\ \exists!\, z \in \mathbb{Z} \quad \text{s.t.} \quad (x, y, z) \in M \end{cases} \tag{2}$$

3 Evaluating Overall Progress and Status

In Γ, define face I, which represents an ideal project progress, as

$$I: \begin{cases} z = (1 - t)\, \sqrt{x^2 + y^2} \\ t = \dfrac{c_2^2}{\sqrt{c_1^2 + 100}} \in \mathbb{R} \qquad (t = \text{const.}) \end{cases} \tag{3}$$

Relocate the points in M and define the resulting group as G, illustrated by the map $f : M \to G$. Define $f(p_k) = g_k = (x'_k,\, y'_k,\, z'_k)$, where

$$\begin{cases} z'_k = z_k \\ \forall\, i, j \in \{1, 2, \ldots, |M|\}, \quad i \neq j \implies f(p_i) \neq f(p_j) \\ \forall\, (x, y, z) \in G, \quad \mathcal{N}(x, y, z) \cap \big(G \setminus \{(x, y, z)\}\big) \neq \emptyset \end{cases} \tag{$*$}$$

where the neighbourhood is defined as

$$\mathcal{N}(x, y, z) = \{(x \pm 1,\, y,\, z),\ (x,\, y \pm 1,\, z)\}.$$

Define

$$E(x, y, z_k) = \sqrt{x^2 + y^2} - \epsilon\, z_k, \qquad \epsilon = \frac{1}{1 - t} = \text{const.}$$

M is relocated to form G such that, for $k \in \{1, 2, \ldots, |M|\}$,

$$g_k = \arg\min_{g \in \mathcal{A}_k} \big|E(x, y, z_k)\big|,$$

where $\mathcal{A}_k$ is the set of dynamically determined available positions for the k-th point, subject to constraint $(*)$.

Since $G \subseteq \Gamma$, let T denote today's date, and define

$$G' = G \cup \{(x, y, T)\} \quad \text{s.t.} \quad |G'| = |\Gamma|, \qquad (x, y, T) \notin G.$$

Solve for the total offset R between I and the points in G':

$$R = \sum_{p \in G'} \big|p_z - z_I(p)\big|, \qquad \text{where } (p_x, p_y, z_I(p)) \in I.$$

The mean

$$\mu = \frac{R}{|\Gamma|}$$

illustrates the overall progress relative to the ideal progress, and

$$\sigma = \sqrt{\frac{1}{|\Gamma|} \sum_{p \in G'} \Big(\big|p_z - z_I(p)\big| - \mu\Big)^2}$$

illustrates the deviation of progress across the project.